

The Evolution of Machine Learning and AI

12-Week Course Structure · Frontier Redesign

Science-inspired AI and the lean-model turn

Award	Certificate in Artificial Intelligence and Machine Learning Engineering (30 ECTS)
Parent programme	Higher Diploma in Science in Computer Science (NFQ Level 8)
Module	The Evolution of Machine Learning and AI (5 ECTS, Semester 1)
Delivery mode	Fully online — live lectures, asynchronous content, self-paced labs
Contact hours	24 lecture + 24 practical + 87 independent learning

Design rationale — why this version is different

The first draft followed the field's textbook chronology and dwelt on classical material. This redesign treats history as a *fast orientation* (one week, not three) and spends the freed time on where the field is actually moving:

1. **Science-inspired AI** — both *AI for science* (AlphaFold 3, GNoME materials discovery, weather models) and *science-grounded architectures* (physics-informed neural networks, neural operators, neural ODEs, diffusion as thermodynamics). The unifying idea: embedding scientific structure as inductive bias makes models more data-efficient and trustworthy.
2. **Scale-free, lean LLMs** — the move beyond brute-force scaling: state-space models (Mamba), linear/attention-free architectures, mixture-of-experts, distillation, low-bit quantization, parameter-efficient fine-tuning, on-device and neuromorphic deployment, and test-time-compute reasoning.
3. **Responsible AI from day one** — bias, fairness and explainability move to **Week 2**, framed as a design constraint that shapes every model built thereafter, rather than a closing topic.

Assessment is two staged pieces: a **mid-module quiz (Week 6)** consolidating the foundations and science strand, and a **final project (Week 12)** on the lean/LLM frontier.

How the semester is shaped

Four **three-week sprints**, each closed by a **reading week (rd)**. Every teaching week carries **2h live online lecture + 2h practical lab**, plus self-paced asynchronous content on Tutors. Sprints 2 and 4 close on an assessment week rather than new content.

Sprint	Weeks	Theme	Closes with
1	1–3	Fast foundations + responsible AI up front	Reading week
2	4–6	Architecture as inductive bias → the science frontier	Week 6 quiz + reading
3	7–9	LLMs, the limits of scale, and the lean turn	Reading week
4	10–12	Systems, frontier trends, final project	Reading / submission

Traceability: indicative content → weeks

All ten descriptor items are still delivered; classical items are compressed, and frontier strands are added on top (flagged in the next table).

#	Indicative content item	Delivered in
1	History of AI/ML: early pioneers, AI winters, resurgence	Week 1 (compressed)

#	Indicative content item	Delivered in
2	Overview of classical machine learning and early neural networks	Week 3 (compressed)
3	Training neural networks: backpropagation, GPU acceleration	Week 3 (compressed)
4	How model structure and connectivity yields families of deep learning models	Week 4
5	Influential breakthroughs and how they improved the state of the art	Threaded — focal Wks 1, 4, 11
6	Open-source and proprietary LLMs (Llama 2, ChatGPT, DeepSeek)	Weeks 7, 8, 9
7	Deep learning frameworks (PyTorch) and runtime environments (Ollama)	Weeks 3, 7, 9
8	Impact of LLMs on software engineering and system design	Weeks 10, 12
9	Ethical AI: bias, fairness, accountability, transparency	Week 2 (early), threaded
10	Emerging trends: federated learning, edge AI, quantum AI	Week 11

Frontier content added beyond the current descriptor

These strands are new and are flagged because, if the programme is to be validated against the descriptor, the descriptor's indicative content (and ideally LOs 2, 3, 7) should be refreshed to name them.

Frontier strand	Delivered in	Examples taught
Science-grounded architectures	Week 5	PINNs, neural operators (FNO, DeepONet), neural ODEs, diffusion as thermodynamics, equivariant/geometric nets, thermodynamics-informed GNNs
AI for scientific discovery	Week 11	AlphaFold 2→3, GNoME materials discovery, protein binder design, weather/climate models, foundation models for science
Scale-free architectures	Week 8	State-space models (S4, Mamba, Mamba-2), RetNet, RWKV, linear/attention-free models, mixture-of-experts, test-time-compute reasoning
Lean / efficient models	Week 9	Distillation (transformer→recurrent), low-bit quantization (incl. 1-bit), pruning, LoRA/QLoRA, small language models, edge/TinyML, neuromorphic/spiking

Sprint 1 — Fast foundations + responsible AI up front

Week 1 — The shape of the field: cycles, breakthroughs, and what the current wave got right

Indicative: 1, 5 · LO1

Lecture (2h). A brisk tour of the evolution — pioneers, two winters, resurgence — framed as *why* the field moves in cycles and *what the modern wave actually got right* (data + compute + architecture). Ends by setting the module thesis: scaling worked, but the frontier is now moving past brute force toward science-grounded and lean systems.

Lab (2h). Use the **AI Evolution Timeline** as a 30-minute macro orientation. Environment setup: Python, Colab, HuggingFace, Ollama, GitHub.

Week 2 — Responsible AI, before we build anything

Indicative: 9 · LO6

Lecture (2h). Bias, fairness, accountability, transparency and explainability introduced as a *design constraint*, not an afterthought. Where bias enters — data, objective, deployment. Documented real-world harms. The EU AI Act and responsible-AI principles as engineering requirements. This lens is applied in every subsequent week.

Lab (2h). Audit a dataset and a model for biased outcomes with the **Bias & Fairness Explorer**; apply explainability tools (SHAP/LIME); see how decision thresholds move fairness metrics.

Week 3 — How machines learn, fast: classical ML, neural nets, training

Indicative: 2, 3, 7 · LO2, LO4

Lecture (2h). Compressed but applied: classical ML (regression, trees, SVMs) → the perceptron and multi-layer networks → backpropagation and gradient descent intuitively → why GPUs and big data unlocked depth. PyTorch introduced as the working framework.

Lab (2h). Train a classical model and a small neural net in PyTorch; use the **Gradient Descent Explorer** to see learning-rate effects, convergence and local minima.

I → Reading week. Quiz revision begins.

Sprint 2 — Architecture as inductive bias → the science frontier

Week 4 — Model families: structure is an assumption about the world

Indicative: 4, 5 · LO2, LO3

Lecture (2h). How connectivity yields capability — CNNs (spatial locality), RNNs/LSTMs (sequence), transformers/attention (relations). The key idea carried into Week 5: *an architecture encodes assumptions about its data*. When those assumptions match reality, models need less data and generalise better.

Lab (2h). Visualise what a CNN learns with the **Convolution / Feature-Map Visualiser**; explore token relations with the **Attention Visualiser**.

Week 5 — Science-inspired AI: building physical law into models

Frontier (extends 4, 5) · LO2

Lecture (2h). Inductive bias drawn from science. Physics-informed neural networks (PINNs) that satisfy governing equations; neural operators (FNO, DeepONet) that learn PDE solution maps; neural ODEs for continuous dynamics; equivariant/geometric deep learning (symmetry); diffusion models understood as non-equilibrium thermodynamics; thermodynamics-informed GNNs. Why physical priors cut data needs and improve reliability.

Lab (2h). Build a small PINN that solves an ODE/PDE (e.g., a damped oscillator or the heat equation) with the **PINN Sandbox**, and compare it against a data-only baseline for accuracy and data efficiency.

Week 6 — Quiz: foundations and the science frontier

Assessment week · consolidates Weeks 1–5 · LO1, LO2, LO4, LO6

Quiz (online, timed). The module's first assessment piece — see Assignment 1 below. Covers the history and cycles of AI, classical ML and neural-network training, model families and frameworks, the science-grounded architectures of Week 5, and the responsible-AI principles from Week 2.

Lab (2h). Structured pre-quiz review clinic: worked examples, concept-check exercises and a live Q&A drawing the first five weeks together; the quiz is sat within or immediately after this session.

I → Reading week. Consolidation; Sprint 3 preview.

Sprint 3 — LLMs, the limits of scale, and the lean turn

Week 7 — Transformers and LLMs: the scaling era and where it strains

Indicative: 6, 7, 5 · LO3, LO4

Lecture (2h). Transformer recap → LLMs: pretraining, fine-tuning, RLHF, tokenisation, emergent ability. Open vs. proprietary models — Llama, ChatGPT, DeepSeek. Then scaling laws *and their limits*: diminishing returns, cost, energy, data ceilings — motivating the lean turn that follows.

Lab (2h). Run LLMs locally with **Ollama**; use the **Tokeniser Explorer**; compare open and proprietary models on the same prompts.

Week 8 — Scale-free LLMs: beyond brute-force attention

Frontier (extends 6, 7) · LO3, LO7

Lecture (2h). The efficiency frontier. The quadratic-attention bottleneck → state-space models (S4, Mamba, Mamba-2), RetNet, RWKV, linear and attention-free architectures with constant memory and linear-time inference. Mixture-of-experts and sparsity for capacity without cost. Test-time-compute / reasoning models that scale inference rather than parameters.

Lab (2h). Run and benchmark a Mamba/SSM model against a comparable transformer; measure latency and memory across sequence lengths with the **Attention-vs-SSM Scaling Visualiser**.

Week 9 — Lean LLMs: small, cheap, private, on-device

Frontier (extends 6, 7) + indicative 10 · LO3, LO4, LO7

Lecture (2h). Making models lean: knowledge distillation (including transformer→recurrent distillation), low-bit quantization (4-bit through 1-bit), pruning, and parameter-efficient fine-tuning (LoRA/QLoRA). Small language models; edge and TinyML deployment; energy use and neuromorphic/spiking approaches. Why lean matters — cost, privacy, sustainability, and access.

Lab (2h). Quantize and LoRA-fine-tune a small model; deploy it under realistic edge constraints; use the **Compression Trade-off Explorer** to weigh size against accuracy.

I → Reading week. Final-project proposals released and reviewed.

Sprint 4 — Systems, frontier trends, final project

Week 10 — LLMs in software engineering and system design

Indicative: 8 · LO5

Lecture (2h). Agentic systems, AI-assisted development, retrieval-augmented generation, embeddings and vector databases — and how to architect systems around *lean local models plus frontier APIs*. The augmentation-not-replacement thesis carried from the Software Engineering certificate.

Lab (2h). Build a RAG application powered by a local lean model; complete a real task with an AI coding agent. Begin final-project build.

Week 11 — Frontier and emerging trends: from AI-for-science to the edge

Indicative: 10 (+ frontier) · LO5, LO7

Lecture (2h). AI as a scientific instrument leads the session — AlphaFold 2→3, GNoME materials discovery, protein binder design, weather/climate and science foundation models, with a critical view that AI tends to *complement* rather than replace experimental work. Then the deployment frontier: federated learning, edge AI, neuromorphic/spiking, quantum ML, and agentic/world models. How the science-inspired and lean currents are converging, and how to weigh substance against hype.

Lab (2h). Explore an AI-predicted scientific structure with the **Protein / Materials Explorer**; run a federated-learning or edge-ML demonstration. Continue final-project build.

Week 12 — Final project: build, showcase, reflect

Assessment week · LO3, LO5, LO7

Showcase (live). The module's culminating assessment — see Assignment 2 below. Students present a working lean/LLM artefact and a critical brief, followed by structured peer review.

Synthesis (closing). A short wrap-up draws the two frontier currents together — grounding models in science, and shrinking them to run anywhere — and reflects on the decade ahead.

I → Reading week. Final-project submission.

Constructive alignment: learning outcomes → weeks → assessment

Module LO	Where addressed	Assessed in
LO1	Trace historical development of AI/ML — Week 1	Quiz (Wk 6)

Module LO	Where addressed	Assessed in
LO2	Characterise training of model families (incl. diffusion/science-grounded) — Weeks 3–5	Quiz (Wk 6)
LO3	Critically evaluate modern LLM architecture and function (incl. lean/scale-free) — Weeks 4, 7–9	Final project (Wk 12)
LO4	Identify tools/frameworks in contemporary AI/ML — Weeks 3, 7, 9	Quiz (Wk 6)
LO5	Analyse LLM influence on software development workflows — Weeks 10, 12	Final project (Wk 12)
LO6	Summarise ethical implications (bias, fairness, explainability) — Week 2 , threaded	Quiz (Wk 6)
LO7	Evaluate emerging trends (science-inspired, lean, edge, federated, quantum) — Weeks 8–9, 11	Final project (Wk 12)

Moving bias to Week 2 means LO6 is taught and consolidated well before the Week 6 quiz.

Assessment strategy (100% continuous)

Assignment 1 — Mid-module quiz (50%, Week 6). *Outcomes 1, 2, 4, 6.* A timed online quiz consolidating the first half: the cycles and breakthroughs of AI; classical ML and neural-network training; model families, frameworks and tools; the science-grounded architectures of Week 5; and the responsible-AI principles introduced in Week 2. A mix of multiple-choice / auto-graded items and short structured responses, with a brief applied component (interpret a model output or a fairness metric).

Assignment 2 — Final project (50%, Week 12 showcase, submitted in the closing reading week). *Outcomes 3, 5, 7.* A critical brief plus a working applied artefact: evaluate a scale-free or lean LLM approach (e.g., an SSM, a distilled/quantized model, or a reasoning model) against a transformer baseline; analyse its impact on a specific software-development or system-design workflow; and assess one emerging trend for a named use case. Assessed via the live Week 12 showcase (presentation + peer review) and the submitted brief and artefact. A proposal milestone is released in the Sprint 3 reading week.

Interactive materials plan (online delivery)

Browser-based, client-side, embeddable in Tutors.dev (markdown + HTML). The plan leads with the frontier interactives.

#	Interactive material	Week	Strand	Serves
1	AI Evolution Timeline — macro orientation to the field's cycles	1	History	LO1
2	Bias & Fairness Explorer — thresholds, metrics, trade-offs (moved early)	2	Responsible AI	LO6
3	Gradient Descent Explorer — learning rate, convergence, minima	3	Foundations	LO2
4	Convolution + Attention Visualiser — what model families “see”	4	Architecture	LO2, LO3
5	PINN Sandbox — solve an ODE/PDE with physics in the loss; physics vs. data baseline	5	Science-inspired	LO2
6	Attention-vs-SSM Scaling Visualiser — quadratic vs. linear cost as sequence length grows	8	Scale-free	LO3

#	Interactive material	Week	Strand	Serves
7	Compression Trade-off Explorer — size vs. accuracy under quantization/distillation	9	Lean models	LO3, LO7
8	Protein / Materials Explorer — interpret AI-predicted scientific structures	11	Science-inspired	LO7

A worked example of item 1 accompanies this document; the **Attention-vs-SSM Scaling Visualiser** (item 6) is the recommended next build, as it embodies the lean-model thesis most directly.

Indicative reading (refreshed)

- Russell, S. & Norvig, P. (2020) *Artificial Intelligence: A Modern Approach*.
- Goodfellow, I., Bengio, Y. & Courville, A. (2016) *Deep Learning*.
- Bronstein, M. et al. *Geometric Deep Learning* (geometric/equivariant architectures).
- Current literature on physics-informed ML, state-space models, and model compression (refreshed each delivery to track the frontier).